NaturArv3D Data Pipeline

Meeting 2026-09-18

Jeppe Klitgaard

s250250@dtu.dk

Outline

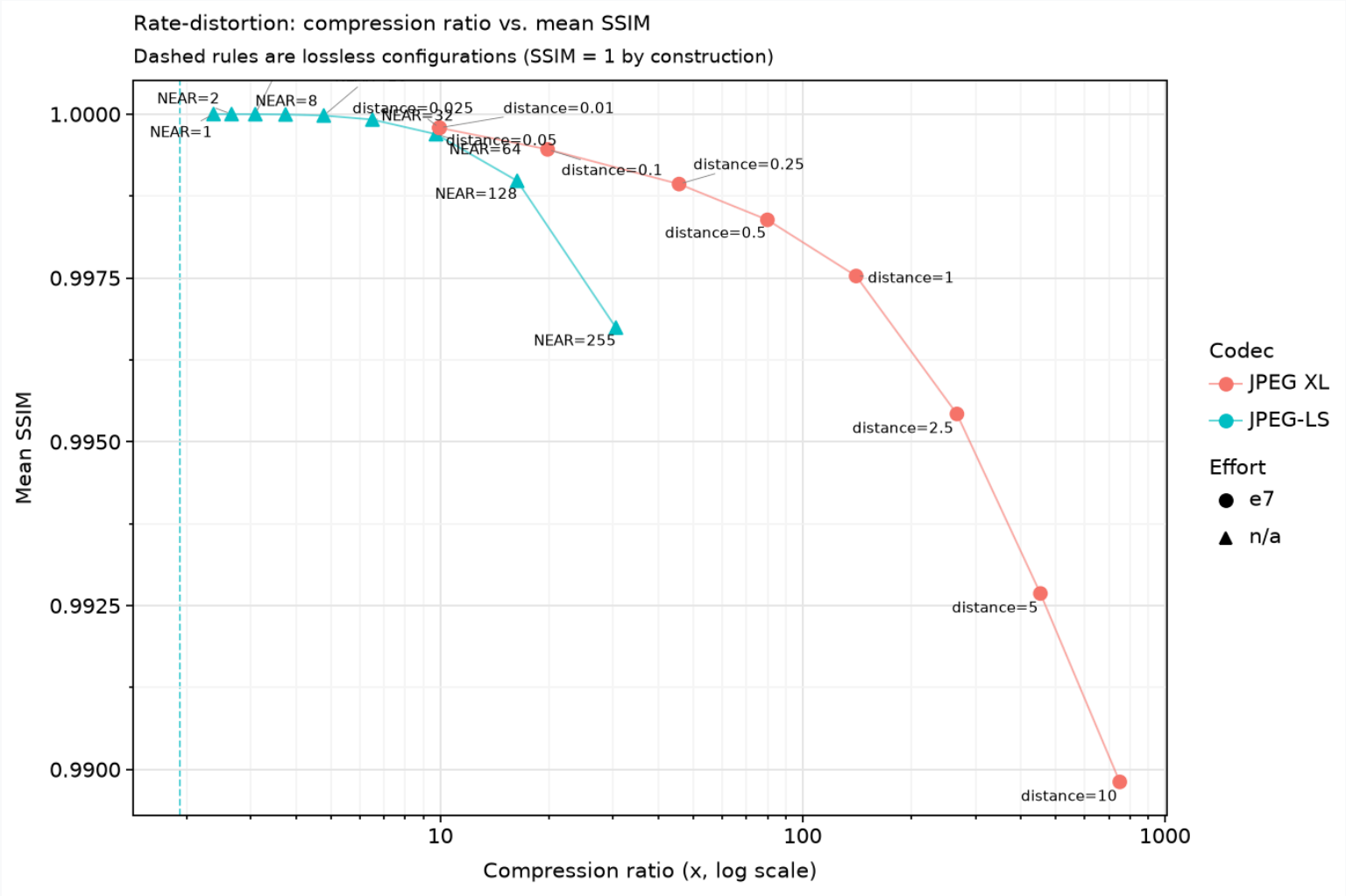
Dataset in Proposed Formats . . . 3
Codec Performance — Fidelity . . . 4
Codec Performance — Throughput . . . 5
Software Support . . . 6
OME-Zarr tooling . . . 7
Summary of options . . . 8
Follow-up: JPEG-XL across compression axis . . . 9
End of slides . . . 10
Format Support . . . 11
Size Considerations . . . 12
Size Consideration Table . . . 13

Dataset in Proposed Formats

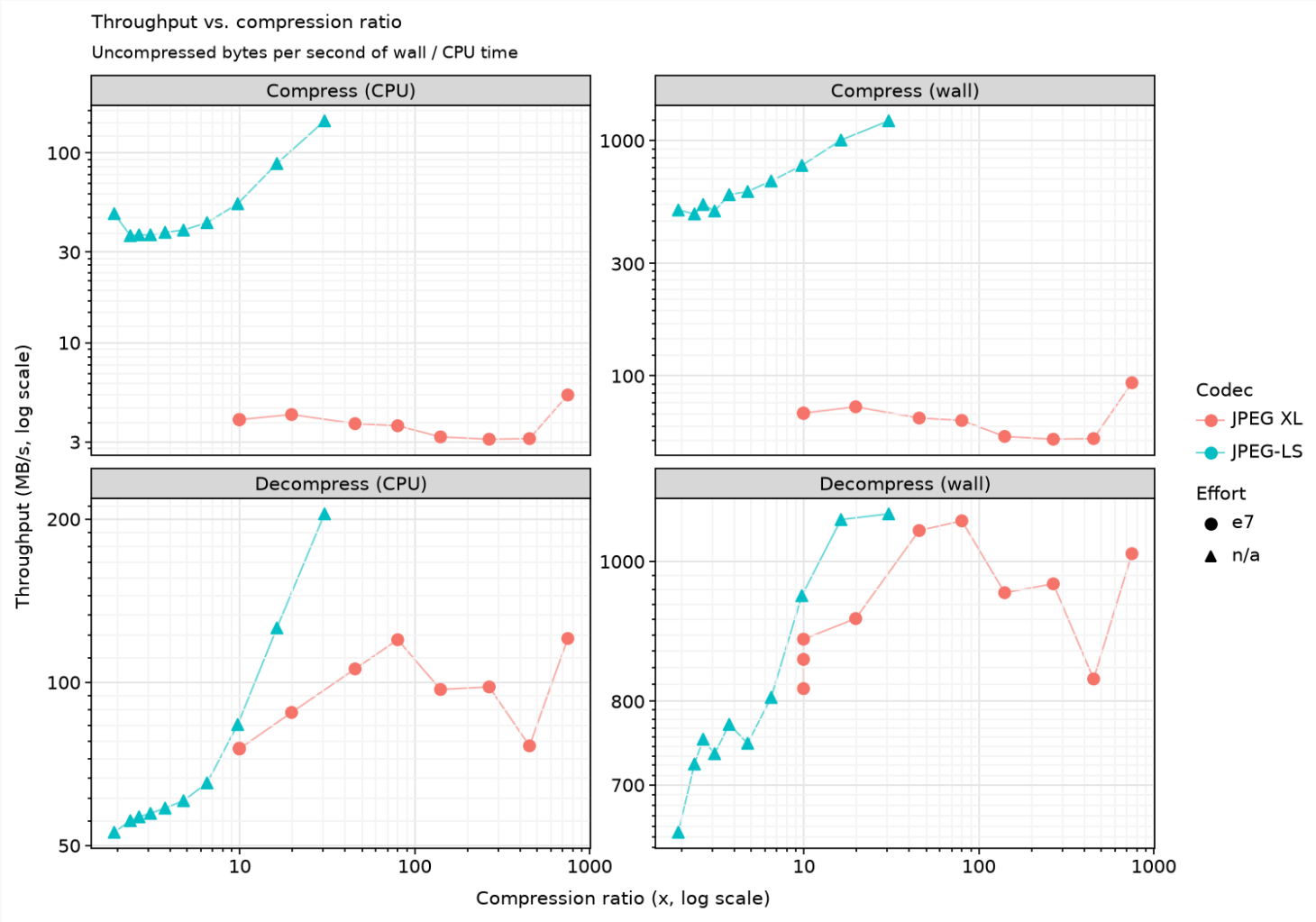
- Agreed last time: make a dataset available in following formats:
  - DICOM
    - No compression
    - JPEG LS
    - ~~JPEG XL~~ *no JPEG XL support in* `pydicom` *yet*
  - OME-Zarr (multi-resolution)
    - No compression
    - JPEG LS
    - JPEG XL
  - Available (for now) here[1]
    - Not sure how to share data in a good way, would like advice

[1] https://mega.nz/folder/pLkyEAKS#fYJAk6cjbiiXGkrguc0qYQ

Codec Performance — Fidelity



Codec Performance — Throughput



Software Support

- DICOMs appear to work in *Napari*, *3D Slicer*, *Fiji* for me
  - No access to other software packages; would be good to test these
- OME-Zarr:
  - Documented in `usage/<VIEWER>` folders in the shared data repository
  - Napari: Works `napari-ome-zarr` plugin, (caveat: compression requires a little trick, for now)
  - Fiji: Works with built-in `BigDataViewer` plugin (maybe still spotty support for codecs)
  - 3D Slicer: Workable by using Python intepreter, see `usage/3D_Slicer`.
  - Likely no support in proprietary software packages, but worth checking

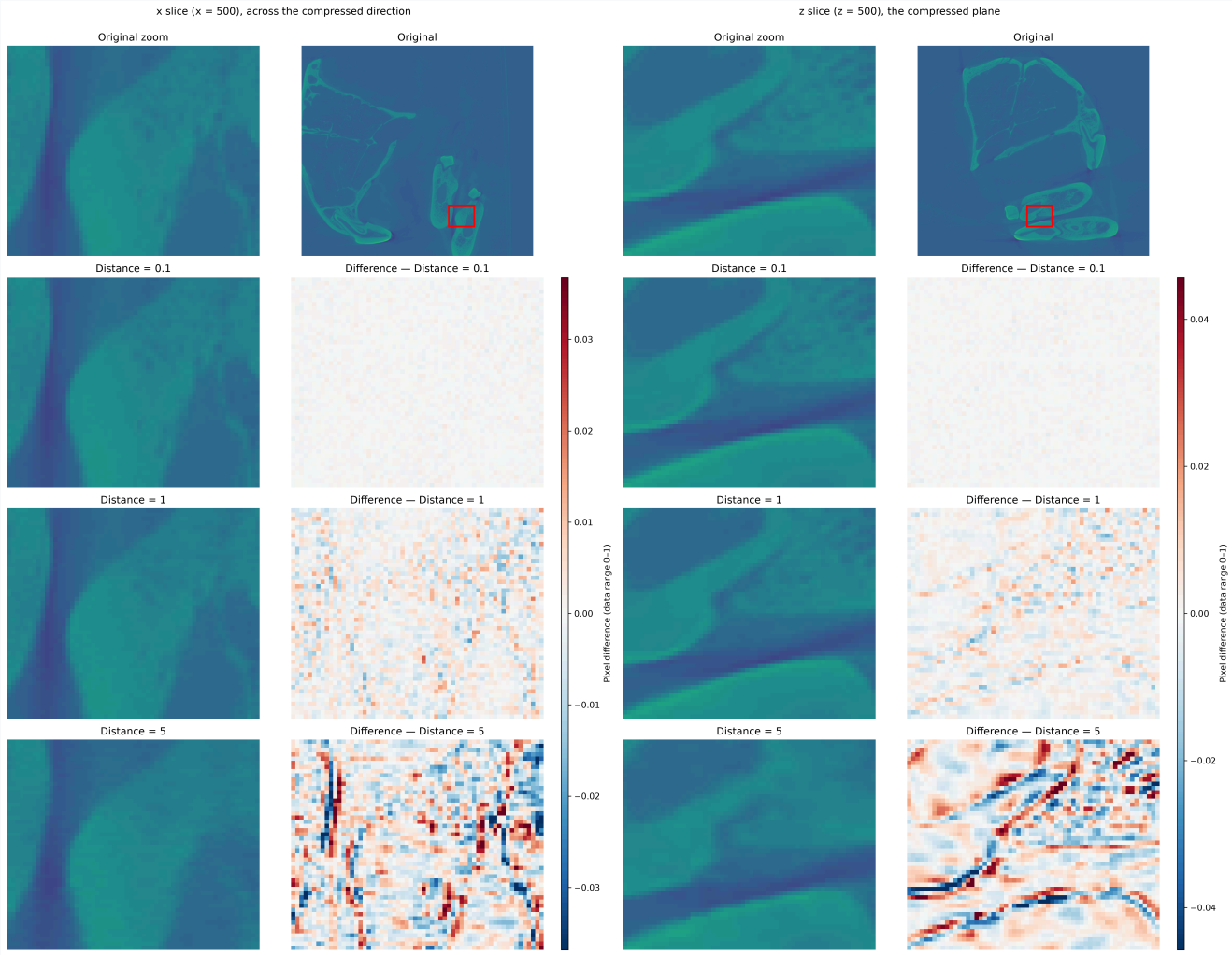
OME-Zarr tooling

- `ome-zarr-py` by the OME (Open Microscopy Environment) organisation is maintained and actively developed
- Possibly open to addition of conversion tool
- Already support for the web-based file serving suggested in earlier meeting
- Thus can serve OME-Zarr datasets directly from object storage (e.g. Backblaze B2 or self-hosted) without downloading the entire dataset first
- Combined with proposed conversion feature: very nice workflow for downloading data, even if locally TIFF stack is required

Summary of options

1. Older, non-lossy-compressed format (TIFF)
2. DICOM with lossy compression (JPEG-LS or JPEG-XL)
3. OME-Zarr with lossy compression (JPEG-LS or JPEG-XL)
   - Combined with easy-to-use tool that can download and convert to TIFF stack locally
   - This also supports multi-resolution, so researcher can download downsampled dataset only
4. ???

Follow-up: JPEG-XL across compression axis



End of slides

Slides that follow this are from previous meetings and are here just for reference.

Format Support

Format	ImageJ/Fiji	Napari	VGSTUDIO	Amira/Avizo	3DSlicer	Dragonfly	Checkpoint	Morphosource	Data Types	Comment
BioFormats	Yes[1]	Yes[1]	No	Yes	No	No	No	No		
TIFF	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	u/i[8/16/32], f16/32	
BigTiff	Yes	Yes[1]	No?	Yes	No?	No?	No?	No?	u/i[8/16/32], f16/32	
JPEG-1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	u8 (u16 LL)	
JPEG-2000	Yes	?	Yes	Yes	No?	No?	No?	Yes?	u1-38	
JPEG-XR	No[2]	?	No?	No[2]	No?	No?	No?	No?	u8/16/32	
JPEG-LS	No?	?	No?	No?	No?	No?	No?	No?	u8/16/32	
JPEG-XL	No	Yes[1]	No	No	No	No	No	No	u8/16/32	
DICOM	Yes	Yes[1]	Yes	Yes	Yes	Yes	Yes	Yes	u8/16	Likely no JXL support?
OME-XML	Yes[1]	Yes[1]	No	Yes	No	No	No	No	u/i[8/16/32], f16/32, …	
OME-TIFF	Yes[1]	Yes[1]	No	Yes	No?	No?	No?	No	u/i[8/16/32], f16/32, …	
OME-ZARR	Yes[1]	Yes[1]	No	Yes[3]	No	No	No	No	u/i[8/16/32], f16/32, …	
HDF5	Yes	Yes[1]	No[4]	Yes[1]	No	Yes?	No	No	u/i[8/16/32], f16/32, …	
PNG	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes?	u8/16	
PPM	Yes	?	Yes	Yes	No?	No?	No	No	u8/16	
RAW	Yes	?	Yes	Yes	Yes	Yes	No	No?		
BMP	Yes	?	Yes	Yes	Yes	Yes	No	Yes	u8	
Analyze/HDR	Yes?	Yes[1]	No	Yes	Yes	Yes	Yes	No	u8, i16/32, f32/64, …	
NifTI	Yes	Yes[1]	No	Yes	Yes	Yes	Yes	No	u/i[8/16/32], f32/64, …	
Other notable					Many		FITS (.fits), Kitware VTK (.vtk)			
Source			E-Mail with *Technical Solutions Export*		3D Slicer Wiki	Dragonfly Manual	Checkpoint	Duke		
Comment	FOSS (GPLv3)	FOSS (BSD-3), Plugin system	Proprietary	Proprietary	FOSS (BSD), Anything via Python		Proprietary	Museum data repository		

[1] With plugin
[2] Supported through `.czi` format
[3] With plugin by Janelia
[4] Does support HDF4

Size Considerations

- Detectors:
  - Large samples: Spectrumlogic, 5606×4800 px, pixel size $50\,\mu\text{m}$
  - Smaller samples: Varex, 3072×3072 px, pixel size $139\,\mu\text{m}$
- *Optimistic* canonical volume: 3072^3 vx $= 28.9$ Gvx (assuming 16-bit)
- Assume 50% (?) crop as an average: 14.4 Gvx
  - 16-bit: $28.8\,\text{GB}$ uncompressed

Cloud storage cost: Backblaze B2 Object Storage[1]:

- Includes 300% monthly egress (data can be downloaded 3 times per month for before additional fees)
- Price is for 5 years of redundant, 'active archive' storage
- 83 USD / TB / year $\Rightarrow$ 0.415 USD / GB $(2.65$ DKK$)$
  - Significantly cheaper than MorphoSource, though this is only for 5 years

[1] https://www.backblaze.com/cloud-storage/pricing, accessed 2026-08-26

Size Consideration Table

Data / Scheme	*Czeibart* 500×500×1000 0% crop		16-bit	2^3 × *Czeibart* 1000×1000×2000 0% crop		16-bit	***Small Detector*** 3072×3072×3072 50% crop		16-bit	*Large Detector* 5606×5606×4800 50% crop		16-bit
Uncompressed	500 MB	250 GB	13.0 TB	4.00 GB	2.00 TB	104 TB	29.0 GB	14.5 TB	756 TB	151 GB	75.4 TB	3.94 PB
CR = 1	11.1 kr	5.54 kkr	289 kkr	88.7 kr	44.3 kkr	2.31 Mkr	643 kr	321 kkr	16.8 Mkr	3.34 kkr	1.67 Mkr	87.3 Mkr
SS = 0.00%	2.07 kr	1.03 kkr	53.9 kkr	16.5 kr	8.27 kkr	431 kkr	120 kr	59.9 kkr	3.13 Mkr	623 kr	312 kkr	16.3 Mkr
	1.33 kr	663 kr	34.6 kkr	10.6 kr	5.30 kkr	277 kkr	76.9 kr	38.4 kkr	2.01 Mkr	400 kr	200 kkr	10.4 Mkr
Lossless	294 MB	147 GB	7.67 TB	2.35 GB	1.18 TB	61.4 TB	17.1 GB	8.53 TB	445 TB	88.7 GB	44.4 TB	2.32 PB
CR = 1.7	6.52 kr	3.26 kkr	170 kkr	52.2 kr	26.1 kkr	1.36 Mkr	378 kr	189 kkr	9.87 Mkr	1.97 kkr	984 kkr	51.3 Mkr
SS = 41.17%	1.22 kr	608 kr	31.7 kkr	9.72 kr	4.86 kkr	254 kkr	70.5 kr	35.2 kkr	1.84 Mkr	367 kr	183 kkr	9.57 Mkr
	780 mkr	390 kr	20.3 kkr	6.24 kr	3.12 kkr	163 kkr	45.2 kr	22.6 kkr	1.18 Mkr	235 kr	118 kkr	6.14 Mkr
Lossy	20.0 MB	10.0 GB	522 GB	160 MB	80.0 GB	4.17 TB	1.16 GB	580 GB	30.3 TB	6.03 GB	3.02 TB	157 TB
CR = 25	443 mkr	222 kr	11.6 kkr	3.55 kr	1.77 kkr	92.6 kkr	25.7 kr	12.9 kkr	671 kkr	134 kr	66.9 kkr	3.49 Mkr
SS = 96.00%	82.7 mkr	41.3 kr	2.16 kkr	661 mkr	331 kr	17.2 kkr	4.79 kr	2.40 kkr	125 kkr	24.9 kr	12.5 kkr	651 kkr
	53.0 mkr	26.5 kr	1.38 kkr	424 mkr	212 kr	11.1 kkr	3.08 kr	1.54 kkr	80.2 kkr	16.0 kr	8.00 kkr	417 kkr
Lossy	10.0 MB	5.00 GB	261 GB	80.0 MB	40.0 GB	2.09 TB	580 MB	290 GB	15.1 TB	3.02 GB	1.51 TB	78.7 TB
CR = 50	222 mkr	111 kr	5.78 kkr	1.77 kr	887 kr	46.3 kkr	12.9 kr	6.43 kkr	335 kkr	66.9 kr	33.4 kkr	1.75 Mkr
SS = 98.00%	41.3 mkr	20.7 kr	1.08 kkr	331 mkr	165 kr	8.62 kkr	2.40 kr	1.20 kkr	62.5 kkr	12.5 kr	6.23 kkr	325 kkr
	26.5 mkr	13.3 kr	692 kr	212 mkr	106 kr	5.53 kkr	1.54 kr	769 kr	40.1 kkr	8.00 kr	4.00 kkr	209 kkr
Lossy	5.00 MB	2.50 GB	130 GB	40.0 MB	20.0 GB	1.04 TB	290 MB	145 GB	7.56 TB	1.51 GB	754 GB	39.4 TB
CR = 100	111 mkr	55.4 kr	2.89 kkr	887 mkr	443 kr	23.1 kkr	6.43 kr	3.21 kkr	168 kkr	33.4 kr	16.7 kkr	873 kkr
SS = 99.00%	20.7 mkr	10.3 kr	539 kr	165 mkr	82.7 kr	4.31 kkr	1.20 kr	599 kr	31.3 kkr	6.23 kr	3.12 kkr	163 kkr
	13.3 mkr	6.63 kr	346 kr	106 mkr	53.0 kr	2.77 kkr	769 mkr	384 kr	20.1 kkr	4.00 kr	2.00 kkr	104 kkr

Size Consideration Table

- Cell rows: data size, storage cost (MorphSource), storage cost (Backblaze B2), storage cost (Cambridge RCS/RFS)
- Cell columns: 1 recon., 1 week of recons. (shaded, 500), 1 year of recons. (26089)
- Units have been SI-ified: mkr = milli-kroner, Mkr = mega-kroner/million, etc.